Mail – Technical Documentation

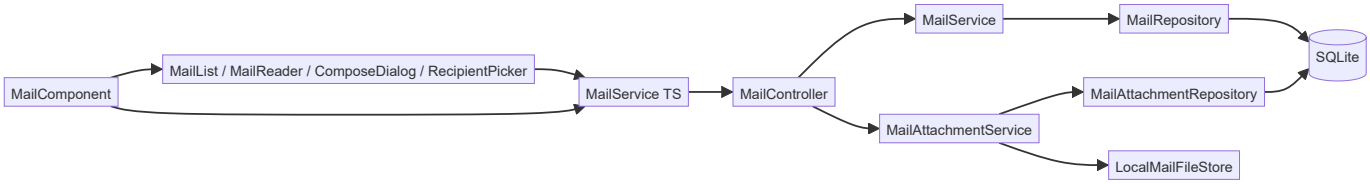
> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Parent Module:** Communication · **Child Module:** 05.2 Mail **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

1. Technical Overview

Internal mail: `MailController` → `MailService` / `MailAttachmentService` → `MailRepository` / `MailAttachmentRepository`. Data lives in `MailMessages`, `MailRecipients`, `MailboxItems` and `MailAttachments`. Each user has their own copy of a mail (`MailboxItem`), so read state, flags and trash are personal. Any query that is not rooted in the caller's own items returns *not found*, never *forbidden*. Attachment bytes are stored on disk by `LocalMailFileStore` under a GUID key.

Every endpoint is under `/api/v1/mail`, requires a JWT (`[Authorize]`) and a permission policy `perm:<key>`, and resolves the caller from the `ClaimTypes.NameIdentifier` claim.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Controller	`TaskFlow.Api/Controllers/Communication/`	`MailController.cs`
Service – Mail	`TaskFlow.Api/Services/Communication/Mail/Messages/`	`IMailService.cs`, `MailService.cs`
Service – Attachments	`TaskFlow.Api/Services/Communication/Mail/Attachments/`	`IMailAttachmentService.cs`, `MailAttachmentService.cs`, `MailAttachmentDownload.cs`
Repository – Mail	`TaskFlow.Api/Repositories/Communication/Mail/Messages/`	`IMailRepository.cs`, `MailRepository.cs`, `MailItemHeader.cs`

Layer	Path	Files
Repository – Attachments	`TaskFlow.Api/Repositories/Communication/Mail/Attachments/`	`IMailAttachmentRepository.cs`, `MailAttachmentRepository.cs`, `AttachmentFile.cs`, `PendingUploads.cs`
DTOs	`TaskFlow.Api/Dtos/Communication/Mail/`	`MailDtos.cs` (`MailView`, `MailComposeMode`, `MailAddressDto`, `MailAttachmentDto`, `MailListItemDto`, `MailMessageDto`, `MailItemDetailDto`, `MailSummaryDto`, `MailUnreadCountDto`, `MailContactDto`, `MailComposeDto`, `SendMailRequest`, `SaveDraftRequest`, `UpdateMailFlagsRequest`, `MailListQuery`)
Validators	`TaskFlow.Api/Dtos/Communication/Mail/Validators/`	`MailRequestValidators.cs` (`SendMailRequestValidator`, `SaveDraftRequestValidator`)
Rules	`TaskFlow.Api/Common/Rules/`	`MailRules.cs`, `EmailPolicy.cs`
Exceptions	`TaskFlow.Api/Common/Exceptions/`	`MailValidationException.cs` (derives `RequestValidationException`), `AppException.cs` (`ConflictException`)
Entities	`TaskFlow.Api/Models/Entities/`	`MailMessage.cs`, `MailRecipient.cs`, `MailboxItem.cs`, `MailAttachment.cs`
Enums	`TaskFlow.Api/Models/Enums/`	`MailFolder.cs`, `MailRecipientKind.cs`
Integration	`TaskFlow.Api/Integrations/Storage/`	`IMailFileStore.cs`, `LocalMailFileStore.cs`
Config	`TaskFlow.Api/Config/`	`MailOptions.cs`
Assistant tools (TFA module)	`TaskFlow.Api/Services/Assistant/Tools/`	`AssistantReadTools.cs` (`FindUserTool`), `AssistantActionTools.cs` (`DraftMailTool`)
Persistence	`TaskFlow.Api/Persistence/`	migration `20261017090431_AddInternalMail` (creates `MailMessages`, `MailRecipients`, `MailboxItems`, `MailAttachments`)

4. Frontend File Structure

Type	Path	Purpose
Route	`src/app/modules/communication/communication.routes.ts`	`/mail` (requires `mail.read`), lazy-loaded with `permissionGuard`
Page component	`src/app/modules/communication/mail/pages/mail.component.ts/.html`	Mail shell: folder rail, counts, URL state (`?folder=&item=`), narrow-screen panes

Type	Path	Purpose
Child components	`src/app/modules/communication/mail/components/`	`mail-list.component` (search, paging, star), `mail-reader.component` (thread, actions, downloads), `compose-dialog.component` (drawer: new/reply/forward/draft, uploads), `recipient-picker.component` (chips, suggestions, validation)
Component models	`src/app/modules/communication/mail/components/compose-dialog.model.ts`, `recipient-picker.model.ts`	`ComposeRequest`, `ComposeResult`, `ComposeField` (server error keys); `RecipientChip`
API service	`src/app/modules/communication/mail/services/mail.service.ts`	All `/api/v1/mail` calls plus the shared `unreadCount` signal for the Mail menu badge
Models	`src/app/modules/communication/mail/models/mail.model.ts`	Mail interfaces and `MAIL_LIMITS`
Utils	`src/app/modules/communication/mail/utils/mail-format.ts`, `linkify.ts`	`MAIL_FOLDERS`, date/size formatting; safe http(s) link splitting (no HTML injection)
Permission constants	`src/app/shared/auth/permissions.ts`	`MailRead`, `MailCompose`, `MailOrganize`, `MailAttachments`
Menu	`src/app/layouts/nav-items.ts`	"Communication Management" → Mail (`mail.read`, badge `mail`)
Shell	`src/app/layouts/shell.component.ts/.html`	Mail badge. The count refreshes on navigation and on a 15 s interval (`SESSION_CHECK_INTERVAL_MS`).

5. API & Logic

5.1 API Endpoints

Route prefix: `api/v1/mail` (class-level `[Authorize]`).

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
1	GET	`/api/v1/mail`	One page of a folder view. Query: `folder` (`MailView`, default Inbox), `search`, `page` (1), `pageSize` (25, max 50)	`mail.read`	Query params (`MailListQuery`)	`PagedResult<MailListItemDto>`
2	GET	`/api/v1/mail/summary`	Folder rail counts	`mail.read`	–	`MailSummaryDto`
3	GET	`/api/v1/mail/unread-count`	Unread Inbox count for the menu badge	`mail.read`	–	`MailUnreadCountDto`
4	GET	`/api/v1/mail/items/{itemId:int}`	Open an item with its conversation; marks it read	`mail.read`	–	`MailItemDetailDto` / `404`

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
5	PATCH	`/api/v1/mail/items/{itemId:int}`	Change personal flags (`isRead` , `isStarred` , `isImportant` ; null = unchanged)	`mail.organize`	`UpdateMailFlagsRequest`	`204` / `404`
6	POST	`/api/v1/mail/items/{itemId:int}/trash`	Move to Trash (sets `TrashedAt` if not set)	`mail.organize`	–	`204` / `404`
7	POST	`/api/v1/mail/items/{itemId:int}/restore`	Restore from Trash	`mail.organize`	–	`204` / `404`
8	DELETE	`/api/v1/mail/items/{itemId:int}`	Delete forever (trashed item) or discard a draft	`mail.organize`	–	`204` / `404` / `409`
9	GET	`/api/v1/mail/items/{itemId:int}/compose?mode=`	Prefilled reply / reply all / forward	`mail.compose`	Query `mode` (`MailComposeMode`)	`MailComposeDto` / `404`
10	POST	`/api/v1/mail/send`	Send a new mail, a reply or forward, or a saved draft	`mail.compose`	`SendMailRequest`	`MailItemDetailDto` (sender's Sent item) / `404`
11	POST	`/api/v1/mail/drafts`	Create a draft	`mail.compose`	`SaveDraftRequest`	`MailItemDetailDto`
12	PUT	`/api/v1/mail/drafts/{messageId:int}`	Update a draft (by **message** id)	`mail.compose`	`SaveDraftRequest`	`MailItemDetailDto` / `404`
13	GET	`/api/v1/mail/contacts?search=`	Up to 10 active users matching name or email	`mail.compose`	Query `search`	`IReadOnlyList<MailContactDto>`
14	POST	`/api/v1/mail/attachments`	Upload one file (multipart `file`). Request limit: 10 MB + 1 MB.	`mail.attachments`	`IFormFile file`	`MailAttachmentDto`
15	GET	`/api/v1/mail/attachments/{attachmentId:int}`	Download as a file (`X-Content-Type-Options: nosniff`)	`mail.attachments`	–	File stream / `404`
16	DELETE	`/api/v1/mail/attachments/{attachmentId:int}`	Remove your own upload that is not yet on any mail	`mail.attachments`	–	`204` / `404`

5.2 Request / Response Samples

```
// POST /api/v1/mail/send
{
  "to": ["anita.k@taskflow.com"],
  "cc": ["ravi.m@taskflow.com"],
  "subject": "Q3 report",
```

```
  "body": "Please review the attached draft.",
  "attachmentIds": [7],
  "draftId": null,
  "parentItemId": null,
  "mode": null
}
```

```
// 200 OK  (MailItemDetailDto, the sender's Sent item)
{
  "itemId": 120,
  "folder": "Sent",
  "isRead": true,
  "isStarred": false,
  "isImportant": false,
  "isTrashed": false,
  "draftTo": null,
  "draftCc": null,
  "message": {
    "messageId": 88,
    "from": { "userId": 4, "name": "Jojo George", "email": "jojo.george@taskflow.com" },
    "to": [{ "userId": 9, "name": "Anita K", "email": "anita.k@taskflow.com" }],
    "cc": [{ "userId": 11, "name": "Ravi M", "email": "ravi.m@taskflow.com" }],
    "subject": "Q3 report",
    "body": "Please review the attached draft.",
    "sentAt": "2026-10-31T10:00:00Z",
    "isSystem": false,
    "taskId": null,
    "attachments": [{ "id": 7, "fileName": "q3.pdf", "contentType": "application/pdf",
"sizeBytes": 245760 }]
  },
  "thread": [ "…same MailMessageDto shape, oldest first…" ]
}
```

```
// 400 Bad Request (MailValidationException -> validation problem details)
{ "status": 400, "errors": { "To": ["Not an active TaskFlow user: someone@taskflow.com."] } }
```

Enums are serialized as strings: `JsonStringEnumConverter` is registered in `Config/CrossCuttingRegistration.cs`. The names and addresses in the sample are illustrative.

5.3 Service Layer Logic

`MailService.SendAsync(int callerId, SendMailRequest request)`

1. If `DraftId` is set, it loads the draft with `GetDraftForUpdateAsync(callerId, draftId)`. If the draft is missing or already sent, it returns `null` (404). Otherwise, if `ParentItemId` is set, it calls `LoadParentAsync`. That returns `null` when the item is not the caller's, and throws `MailValidationException("Mode", "Mails from TaskFlow cannot be replied to. Forward it instead.")` for Reply or ReplyAll on a system mail.
2. Subject and body come from the request, falling back to the draft. If both are empty it throws "Write a subject or a message before sending."

3. `ResolveRecipientsAsync`: normalises the addresses (trim, lower-case, remove duplicates) and removes Cc entries already in To. There must be 1 to 50 addresses. `GetActiveUserIdsByEmailAsync` then resolves them, and any unknown address produces `Not an active TaskFlow user: …` per field (To/Cc).
4. A draft sent with only its id keeps its saved `DraftTo` / `DraftCc` and its attachments.
5. `PlanAttachmentsAsync` validates everything before writing. The ids must be the caller's own uploads, either unattached or already on this draft. The total must be 5 files or fewer and 25 MB or less, counting forwarded copies. Uploads previously on the draft but no longer listed are detached.
6. `InTransactionAsync` (with `DetachAll` in a `finally` block): creates or reuses the `MailMessage` (`ConversationId` is the parent's id or a new GUID, `ParentMessageId`, `SentAt = now`, draft fields cleared). It applies the attachments; forwarded files become new `MailAttachment` rows that share the same `StorageKey`. It adds `MailRecipient` rows (To/Cc), one Sent `MailboxItem` for the sender (`IsRead = true`) and one Inbox `MailboxItem` per recipient, and removes the draft's Drafts item.
7. Returns `BuildDetailAsync(callerId, sentItemId)`: the item header plus the conversation (messages in the caller's mailbox that are not drafts), ordered by `SentAt`.

`CreateDraftAsync` / **`UpdateDraftAsync`**: validate the parent (on create) and the attachments. `ApplyDraftFields` stores the trimmed subject, the body, and raw `DraftTo` / `DraftCc` joined with ", ". Create adds a Drafts `MailboxItem` (`IsRead = true`). Update sets `SortAt = now`.

`ListAsync`: page at least 1, page size clamped to 1–50, then `MailRepository.ListAsync`. **`GetSummaryAsync`**: unread counts for Inbox, Starred and Important, plus the total Drafts count. **`GetUnreadCountAsync`**: unread Inbox count.

`OpenAsync`: `GetItemForUpdateAsync`. If the item is unread, sets `IsRead = true` and saves, then returns the detail.

`UpdateFlagsAsync` / **`TrashAsync`** (**`TrashedAt ??= now`**) / **`RestoreAsync`** (**`TrashedAt = null`**) each return `false` (404) when the item is not the caller's.

`DeleteForeverAsync`: if the item is neither trashed nor in Drafts it throws `ConflictException("Move the mail to Trash before deleting it forever.")` (409). Otherwise it sets `PurgedAt = now`, a soft purge that keeps the row.

`GetComposeAsync(callerId, itemId, mode)`:

- **Reply:** To = the sender, or the original To if the caller sent it.
- **ReplyAll:** the sender plus To and Cc, minus the caller.
- **Forward:** no recipients and `ForwardedAttachments` = the original attachments.
- **Subject:** `MailRules.WithPrefix("Re: " / "Fwd: ")`, max 200.
- **Body:** a quoted `>` reply with "On {dd MMM yyyy, HH:mm}, {name} wrote:", or a "---------- Forwarded message ----------" header. Both use the application timezone and are truncated to 20,000 characters.

`GetContactsAsync(search)`: an empty or blank term returns `[]`. Otherwise it calls `SearchContactsAsync(term, 10)`.

`MailAttachmentService.UploadAsync(callerId, stream, fileName)`

1. `MailRules.SanitizeFileName`: keeps the last path segment, removes invalid and control characters, allows at most 255 characters, and falls back to "file".
2. Rejects `BlockedExtensions`.
3. Checks pending uploads (`GetPendingUploadsAsync`): at most 20 files and 100 MB per user.
4. `IMailFileStore.SaveAsync(stream, min(10 MB, remaining room))`. This throws "A file can be at most {n} MB." when the file is over the limit, and that is re-worded when it was the pending room that was exceeded. A size of 0 gives "The file is empty."

5. Content type comes from the extension allow-list (`SafeTypes` : images, pdf, txt, csv, Office formats, zip), otherwise `application/octet-stream` . It is never taken from the client.
6. Adds a `MailAttachment` with no `MessageId` (pending) and saves.

`OpenAsync(callerId, attachmentId)` : `GetDownloadableAsync` succeeds when the caller is the uploader, or the message is in one of the caller's mailbox items that is not purged. It then calls `_store.OpenRead(key)` . A missing record or file returns 404.

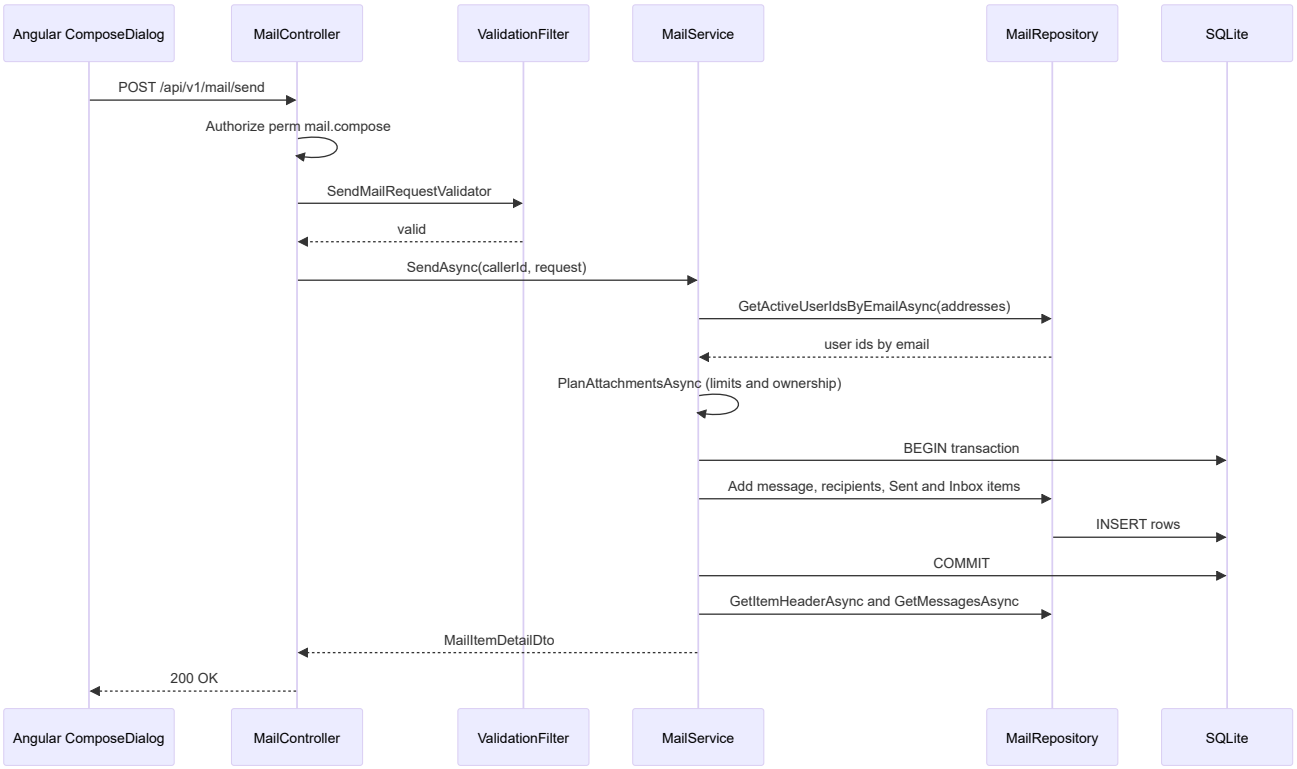
`DeleteAsync` : only the caller's own pending upload. It removes the record only, because the stored file may be shared by a forwarded copy.

5.4 Repository Layer

Method	Description	Query Notes
`Mine(ownerId)` (private)	Base of every mailbox query	`MailboxItems.Where(OwnerUserId == owner && PurgedAt == null)`
`ListAsync(owner, view, search, page, size)`	Folder page	`InView` filter (Inbox/Sent/Drafts by `Folder` and not trashed; Starred/Important by flag, excluding Drafts; Trash = `TrashedAt != null`). Search: escaped `LIKE` on Subject, Body, DraftTo, the "TaskFlow" system sender, and sender or recipient FullName/Email (`IgnoreQueryFilters` so deleted users still match). Order by `SortAt` desc, `Id` desc. Body preview uses the first 300 characters, reduced to 120 by `MailRules.Preview` .
`CountAsync(owner, view, unreadOnly)`	Folder counts	Same `InView` filter
`GetItemForUpdateAsync` / `GetItemHeaderAsync`	Own item (tracked / projected `MailItemHeader`)	–
`GetDraftForUpdateAsync(owner, draftMessageId)`	Own Drafts item by **message** id	`Include(Message)`
`GetMessageOfItemAsync` / `GetMessageIdOfItemAsync`	Parent message, excluding drafts	–
`GetConversationMessageIdsAsync(owner, conversationId)`	Thread in the caller's mailbox	Excludes Drafts, `Distinct`
`GetMessagesAsync(ids)`	Full `MailMessageDto` list	Loads people with `IgnoreQueryFilters` . The system sender is shown as `MailAddressDto(null, "TaskFlow", "")` , and a user id that is not found as "Unknown user".

Method	Description	Query Notes
`GetActiveUserIdsByEmailAsync(emails)`	Resolve recipients	`IsActive` and lower-case email match. The global query filter already hides deleted users.
`SearchContactsAsync(term, max)`	Recipient picker	Active users, `LIKE` on FullName or Email, order by FullName, `Take(max)`
`AddRecipient` / `AddMailboxItem` / `RemoveMailboxItem` / `DiscardStagedMail`	Change-tracker helpers	`DiscardStagedMail` detaches mail entities that were added but not saved (used by `TaskCommentService` on failure)
`MailAttachmentRepository.GetOwnAvailableForUpdateAsync(uploader, ids, messageId)`	Attachments the caller may use	Uploader matches, and `MessageId` is null or the same message
`GetOnMessageAsync` / `GetIdsOnMessageAsync` / `GetOthersOnMessageForUpdateAsync`	Attachment sets on a message	–
`GetPendingUploadsAsync(uploader)`	Count and bytes of unattached uploads	`GroupBy` then `Count` / `Sum` into `PendingUploads`
`GetDownloadableAsync(caller, id)`	Download permission check	Uploader, **or** a `MailboxItems` row for the message owned by the caller with `PurgedAt == null`
`GetOwnPendingForUpdateAsync(uploader, id)`	Deletable upload	Uploader matches and `MessageId == null`

5.5 Sequence Diagram



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6. Database Design

Column types are given as C# type → SQLite storage. Enums that use `HasConversion<string>()` are stored as TEXT.

Table	Column	Type	Constraints	Description
`MailMessages`	`Id`	int → INTEGER	PK	–
`MailMessages`	`ConversationId`	Guid → TEXT	Index	Thread key
`MailMessages`	`ParentMessageId`	int? → INTEGER	FK → `MailMessages.Id`	Reply/forward source
`MailMessages`	`SenderUserId`	int? → INTEGER	FK → `Users.Id`, nullable	Null for system mail
`MailMessages`	`IsSystem`	bool → INTEGER	–	TaskFlow sender
`MailMessages`	`Subject`	string → TEXT(200)	Required	–
`MailMessages`	`Body`	string → TEXT(20000)	Required	Plain text
`MailMessages`	`TaskId`	int? → INTEGER	FK → `Tasks.Id`	"Open the task" link
`MailMessages`	`ReminderHistoryId`	int? → INTEGER	FK → `ReminderHistory.Id`; **unique filtered index** (`IS NOT NULL`)	One mail per reminder

Table	Column	Type	Constraints	Description
`MailMessages`	`DraftTo` / `DraftCc`	string? → TEXT(4000)	nullable	Raw draft recipients
`MailMessages`	`CreatedAt` / `UpdatedAt` / `SentAt`	DateTime → TEXT (`SentAt` nullable)	–	`SentAt` null = draft
`MailRecipients`	`Id`	int → INTEGER	PK	–
`MailRecipients`	`MessageId`	int → INTEGER	FK → `MailMessages.Id`	–
`MailRecipients`	`UserId`	int → INTEGER	FK → `Users.Id`	–
`MailRecipients`	`Kind`	`MailRecipientKind` → TEXT(8)	Not Null	To / Cc
`MailRecipients`	–	–	**Unique** `(MessageId, UserId)`	–
`MailboxItems`	`Id`	int → INTEGER	PK	Item id used by the API
`MailboxItems`	`MessageId`	int → INTEGER	FK → `MailMessages.Id`	–
`MailboxItems`	`OwnerUserId`	int → INTEGER	FK → `Users.Id`	Mailbox owner
`MailboxItems`	`Folder`	`MailFolder` → TEXT(16)	Not Null	Inbox / Sent / Drafts
`MailboxItems`	`IsRead` / `IsStarred` / `IsImportant`	bool → INTEGER	–	Personal flags
`MailboxItems`	`TrashedAt` / `PurgedAt`	DateTime? → TEXT	nullable	Trash / deleted forever (soft)
`MailboxItems`	`SortAt`	DateTime → TEXT	–	Sent time or last draft save
`MailboxItems`	–	–	**Unique** `(MessageId, OwnerUserId, Folder)` ; Index `(OwnerUserId, Folder, PurgedAt, TrashedAt, SortAt)`	–
`MailAttachments`	`Id`	int → INTEGER	PK	–

Table	Column	Type	Constraints	Description
`MailAttachments`	`MessageId`	int? → INTEGER	FK → `MailMessages.Id`, Index	Null = pending upload
`MailAttachments`	`UploadedByUserId`	int → INTEGER	FK → `Users.Id`	–
`MailAttachments`	`FileName`	string → TEXT(255)	Required	Sanitised
`MailAttachments`	`ContentType`	string → TEXT(128)	Required	From the extension allow-list
`MailAttachments`	`SizeBytes`	long → INTEGER	–	–
`MailAttachments`	`StorageKey`	Guid → TEXT	–	File name in the store; shared by forwarded copies
`MailAttachments`	`CreatedAt`	DateTime → TEXT	–	–

Relationships

- `MailMessages.ParentMessageId → MailMessages.Id` (FK, Restrict)
- `MailMessages.SenderUserId → Users.Id` (FK, Restrict)
- `MailMessages.TaskId → Tasks.Id` (FK, Restrict)
- `MailMessages.ReminderHistoryId → ReminderHistory.Id` (FK, Restrict) – `ReminderHistory` is owned by Notifications (05.1)
- `MailRecipients.MessageId → MailMessages.Id` (FK, Cascade); `MailRecipients.UserId → Users.Id` (FK, Restrict)
- `MailboxItems.MessageId → MailMessages.Id` (FK, Cascade); `MailboxItems.OwnerUserId → Users.Id` (FK, Restrict)
- `MailAttachments.MessageId → MailMessages.Id` (FK, Cascade); `MailAttachments.UploadedByUserId → Users.Id` (FK, Restrict)
- Reads: `Users` (`FullName`, `Email`, `IsActive`) for recipients and contacts; `ApplicationSettings` key `Application.Timezone` for quoted dates.

7. Security & Authorization

- **Authentication and authorization:** as described in the parent document (`[Authorize]`, `perm:<key>` policies, `DeniedActionAuditHandler`).
- **Data scope:** always **Own**. Mail queries start from `MailRepository.Mine(caller)`, and attachment downloads require the caller to be the uploader or to have a non-purged mailbox item for the message. A record belonging to someone else returns **404, never 403**. The one exception is `GET /mail/contacts`, which searches the whole directory of active users.

- **Content safety:** attachments are always served as downloads with a server-chosen content type and `X-Content-Type-Options: nosniff`. Mail bodies are plain text, and the frontend `linkify` turns only http(s) URLs into links through Angular bindings. Upload request size is capped at 11 MB.
- **Frontend:** `permissionGuard` on `/mail`. The menu entry and badge are built from `has(Permissions.MailRead)`. Components hide buttons with `canCompose()` and `canOrganize()` computed from `AuthService.hasPermission`.

Permission Key	Endpoint(s)	UI Element Controlled
`mail.read`	`GET /api/v1/mail`, `/summary`, `/unread-count`, `/items/{id}`	Menu "Mail" and badge, route `/mail`, folder rail, list, reader
`mail.compose`	`GET /items/{id}/compose`, `POST /send`, `POST /drafts`, `PUT /drafts/{id}`, `GET /contacts`	Compose buttons (rail and narrow), Reply / Reply all / Forward
`mail.organize`	`PATCH /items/{id}`, `POST /items/{id}/trash`, `POST /items/{id}/restore`, `DELETE /items/{id}`	Star toggle in list, Star / Important / Mark unread / Trash / Restore / Delete forever in reader. **Draft Discard** in the compose panel calls `DELETE /items/{id}` but is not gated by this key.
`mail.attachments`	`POST /attachments`, `GET /attachments/{id}`, `DELETE /attachments/{id}`	Attach button, attachment chips, download buttons (**none of them gated in the UI**)

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
Subject over 200 / body over 20,000 characters	`SendMailRequestValidator`, `SaveDraftRequestValidator`	400	FluentValidation default maximum-length message
No subject and no body (new send)	Validator and `MailService`	400	Write a subject or a message before sending.
Recipient count not 1–50 (new send)	Validator	400	Add between 1 and 50 recipients.
Address shape (needs @, no spaces, at most 320 characters)	Validator	400	'{PropertyValue}' is not an email address.
More than 5 distinct attachment ids	Validator and `MailService`	400	A mail can have at most 5 attachments.
`Mode` missing or invalid when `ParentItemId` is set	Validator	400	FluentValidation default NotNull/IsInEnum message
Draft: more than 50 To/Cc, or address over 320 characters	`SaveDraftRequestValidator`	400	FluentValidation default message

Rule	Layer	HTTP Code	Error Message
No recipients after normalising	`MailService`	400	Add at least one recipient.
Over 50 recipients	`MailService`	400	A mail can have at most 50 recipients.
Unknown or inactive recipient	`MailService`	400	Not an active TaskFlow user: {addresses}. (field `To` or `Cc`)
Attachment not the caller's or already used	`MailService`	400	One or more attachments are not available.
Attachments over 25 MB total	`MailService`	400	Attachments on one mail can total at most 25 MB.
Reply / Reply all to a system mail	`MailService`	400	Mails from TaskFlow cannot be replied to. Forward it instead.
Delete forever on an item that is not trashed and not a draft	`MailService` (`ConflictException`)	409	Move the mail to Trash before deleting it forever.
No file in upload	`MailController`	400	Choose a file to upload.
Blocked extension	`MailAttachmentService`	400	Files of type {ext} are not allowed.
File over 10 MB	`LocalMailFileStore`	400	A file can be at most 10 MB.
Empty file	`MailAttachmentService`	400	The file is empty.
More than 20 pending uploads	`MailAttachmentService`	400	You have {n} files waiting to be attached. Send or discard a mail before uploading more.
Pending uploads at or over 100 MB	`MailAttachmentService`	400	Your files waiting to be attached already total 100 MB. Send or discard a mail before uploading more. / This file would take your files waiting to be attached past 100 MB. Send or discard a mail first.

`MailValidationException` derives from `RequestValidationException`, so `GlobalExceptionHandler` turns it into a 400 validation problem with an `errors` map keyed by field (`To`, `Cc`, `Body`, `AttachmentIds`, `Mode`, `File`). The compose panel maps these keys onto its field errors (`ComposeField` in `compose-dialog.model.ts`). The shared 401 / 403 / 404 / 409 concurrency / 500 rules are in the parent document §8.

9. Integrations / Side Effects

- **Reminder mails (from Notifications 05.1):** `IsSystem = true`, linked to `ReminderHistoryId`, sent when the `Notification.MailEnabled` setting is on. That setting is the "Also deliver reminders to Mail" toggle on the Reminder Settings screen.
- **Comment mails** from `TaskCommentService.AddInboxMail` (`IsSystem = true`, subject "Comment {verb}: {title}") follow the same setting.
- **Reassignment mails:** `TaskService` calls `IMailService.SendAsync` **as the actor** (not as a system mail), with subject "Task reassigned: {title}".

- **File storage:** `LocalMailFileStore` writes each attachment once under `{Storage:Root}/{Mail:AttachmentsPath}/{guid}`. Files are never changed and are never deleted by `DeleteAsync`.
- **AI Assistant:** `FindUserTool` (read tool) calls `IMailService.GetContactsAsync`. `DraftMailTool` ("draft_mail") prepares a `SendMailRequest`, validates it with `IValidator<SendMailRequest>`, and records a `SendMail` pending action that the user must confirm. The help articles `mail-folders.md` and `compose-mail.md` in `Services/Assistant/Knowledge/` describe this child module.
- **Activity Log:** Mail writes no task activity entries.

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`Mail:AttachmentsPath`	Attachment folder. A relative path resolves against `Storage:Root`.	`mail-attachments`

`Mail:AppBaseUrl` (also in `MailOptions`) is used only by reminder mails and is documented in `05.1-Notifications.md` §10. The shared keys (`Storage:Root`, `Application:Timezone`, `ConnectionStrings:Default`, `Notification.MailEnabled`, `Application.Timezone`) are in the parent document §10.

11. Testing

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Communication/Mail/Messages/MailSendTests.cs`	Send delivers to To and Cc Inboxes (unread) and the sender's Sent folder; an address that is not an active TaskFlow user (external, unknown, deactivated or deleted) rejects the whole send with a field error and stores nothing; a recipient and some content are required; mailing yourself lands in Sent and Inbox; addresses are case-insensitive and a To/Cc duplicate is delivered once as To; `SendMailRequestValidator` limits (subject, body, 50 recipients, address shape, 5 attachments) and a draft send needs no other fields
Unit	`tests/TaskFlow.UnitTests/Communication/Mail/Messages/MailDraftAndThreadTests.cs`	Drafts are private to the author, may hold unfinished recipients, can be edited and sent (moving to Sent) and keep the send rules; only the author can edit, send or discard a draft; Reply and Reply all prefill recipients, one `Re:` prefix and a quote, also from Sent mail; Forward has no recipients, a `Fwd:` quote and carries attachments without copying files; the conversation shows only my mails of the thread, oldest first; system mail cannot be replied to but can be forwarded
Unit	`tests/TaskFlow.UnitTests/Communication/Mail/Messages/MailboxTests.cs`	Inbox, Sent, Starred, Important and Trash views; list rows newest first with flags and a one-line preview; paging; read state and unread count per recipient, and the folder summary; another user's item is not found even for an Administrator; trash, restore and delete forever affect only my copy (delete forever outside Trash is a conflict); flags are personal; search over subject, body and people within my mailbox and folder (`%` and `_` are literal), skipping Trash unless inside it; contacts return up to 10 active users
Unit	`tests/TaskFlow.UnitTests/Communication/Mail/Messages/MailTestHarness.cs`	Shared helper, not a test class: an in-memory database with four users, a `MailService` on a controllable clock, and send and folder helpers

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Communication/Mail/Attachments/MailAttachmentTests.cs`	An uploaded file sent with a mail can be downloaded by recipients; outsiders cannot download and unsent uploads stay private; a recipient loses access after delete forever; files over 10 MB, blocked types (`.exe` , `.ps1` , `.js`) and empty files are rejected and nothing is stored; path tricks in the name are harmless; per-mail count and total-size limits; another user's upload cannot be attached; a draft keeps its attachments and removing one detaches it; pending uploads are capped by count and by 100 MB total
Unit	`tests/TaskFlow.UnitTests/Common/EmailDomainTests.cs`	`EmailPolicy` : only `@taskflow.com` addresses with a valid local part (64 characters at most) pass, and `Normalize` trims and lowercases. The same file also tests user email validation and uniqueness for User Management
Integration	`tests/TaskFlow.IntegrationTests/Communication/Mail/MailEndpointsTests.cs`	`POST /api/v1/mail/send` updates the sender's Sent and the recipient's Inbox and `unread-count` ; an external address returns 400 and delivers nothing; another user's item returns 404 for GET, PATCH and trash, even for an Administrator; no token returns 401; Reply all compose and send keep one thread; draft create, update and send; star, search, trash, restore and delete forever (409 outside Trash); attachment upload, blocked type 400, download as `attachment` with `nosniff` , 404 for outsiders; contacts list only active users and are open to Employees
Frontend unit	`src/app/modules/communication/mail/services/mail.service.spec.ts`	`MailService` HTTP calls: folder list with search and paging; badge unread count that ignores failures; open, flags, trash, restore and delete; send then refresh the badge; drafts and compose prefill; multipart upload and blob download; contact search
Frontend unit	`src/app/modules/communication/mail/utils/linkify.spec.ts`	`linkify` turns only `http(s)` addresses into link parts, keeps surrounding text and leaves closing punctuation out; `sameOriginPath` returns the in-app path for this origin and `null` otherwise
Frontend unit	`src/app/modules/communication/mail/pages/mail.component.spec.ts`	Folder list with unread and draft counts; folder and opened mail come from the URL; a mail opens in the reader but a draft in the composer; list and counts reload after send; one pane at a time on narrow screens, all panes on wide screens
Frontend unit	`src/app/modules/communication/mail/components/mail-list.component.spec.ts`	Loads the folder with unread rows in bold; shows recipients in Sent and "TaskFlow" for system mail; reloads on folder change; opening a row marks it read; star toggle without opening; debounced search; empty state
Frontend unit	`src/app/modules/communication/mail/components/mail-reader.component.spec.ts`	Opens the mail with its conversation and refreshes the badge; body links (same-origin links open in the app, others in a new tab) and no HTML from the body; Reply, Reply all and Forward for a person's mail, only Forward and the task link for system mail; star, important, trash, restore, delete forever, mark unread; attachment download

Test Type	File / Location	Covers
Frontend unit	`src/app/modules/communication/mail/components/compose-dialog.component.spec.ts`	Sends a new mail; blocks sending while a recipient is not a TaskFlow user or nothing is filled in; Reply all prefill sent in the conversation; forwarded files shown; continue, save and discard drafts; oversized or blocked files refused before upload; upload and remove files; server field errors shown next to fields; an address still being typed is included on Send and kept on Save draft
Frontend unit	`src/app/modules/communication/mail/components/recipient-picker.component.spec.ts`	Suggests TaskFlow users while typing; accepts a typed TaskFlow address and flags one that is not; checks chips passed in from outside once each; removes chips by button or Backspace; commits typed text as a chip on demand

The cross-module comment test, which also sends Inbox mail, is listed in the parent document §11.

12. Known Limitations / Gaps found

- "Delete forever" is a soft purge (`PurgedAt`). Rows and attachment files are never physically removed, and there is no clean-up job for orphaned files or abandoned uploads.

Gaps found

1. **`mail.attachments` is not checked in the UI.** The Attach button, chip removal and download buttons are shown regardless of the key, so users without it get 403.
2. **Draft "Discard" is not gated.** In `compose-dialog.component.ts` it calls `DELETE /api/v1/mail/items/{id}` , which requires `mail.organize` , but it is only implicitly gated by `mail.compose` (the panel opening). A role with compose but not organize cannot discard a saved draft.
3. **Contacts endpoint exposes the directory.** `GET /api/v1/mail/contacts` returns the organization-wide active-user directory (name and email) to anyone with `mail.compose` .
4. **`IMailService.DeleteForeverAsync` has an inaccurate doc comment.** It says it throws `InvalidOperationException` , but the implementation throws `ConflictException` (409).

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